Snapdeal

Chapter-1

Introduction

Snapdeal is one of India's leading e-commerce platforms, founded in **2010** by Kunal Bahl and Rohit Bansal. Headquartered in New Delhi, Snapdeal began as a daily deals website and later evolved into a full-fledged online marketplace. It connects millions of customers with thousands of sellers across India, offering a wide range of products, including fashion, electronics, home essentials, beauty products, books, toys, and accessories. The platform aims to provide quality products at affordable prices while making online shopping convenient and accessible.

Snapdeal operates on a marketplace model, allowing third-party sellers to list and sell their products through its website and mobile application. The platform offers secure payment options, Cash on Delivery (COD), easy returns, order tracking, and regular discounts to enhance the customer shopping experience. By focusing on affordable products and expanding its reach to Tier-2 and Tier-3 cities, Snapdeal has become an important player in India's e-commerce industry, supporting both consumers and businesses through a reliable and user-friendly digital shopping platform.

Objective

The primary objective of this project is to design and develop an efficient database management system for the Snapdeal e-commerce platform. The database is intended to store, organize, and manage essential business information such as customer details, seller information, product catalog, categories, orders, payments, deliveries, and reviews. It aims to ensure accurate data storage, maintain data integrity, and support smooth business operations.

The project also focuses on improving data retrieval, reducing redundancy, and enabling efficient transaction processing. By creating a well-structured relational database, the system helps manage online shopping activities effectively, supports decision-making through accurate data, and provides a scalable foundation for handling the growing number of customers, products, and orders on the Snapdeal platform.

Key Objectives

- To design a structured and normalized database for the Snapdeal e-commerce platform.
- To store and manage customer, seller, product, category, and order information efficiently.

- To maintain accurate records of payments, shipments, and order status.
- To reduce data redundancy and ensure data integrity through proper database design.
- To establish relationships between database tables using primary and foreign keys.
- To enable fast and accurate data retrieval using SQL queries.
- To support secure storage and management of customer and business data.
- To improve the efficiency of online shopping operations and transaction processing.
- To generate useful reports for analyzing sales, customers, products, and orders.
- To build a scalable database that can accommodate future business growth and increasing data volume.

Problem Statement

As an e-commerce platform grows, managing large volumes of data related to customers, sellers, products, orders, payments, deliveries, and reviews becomes increasingly complex. Without a well-designed database, information may become inconsistent, duplicated, or difficult to retrieve, leading to delays in order processing, inventory management, and customer service. These issues can negatively affect business operations and customer satisfaction.

The purpose of this project is to design a structured and efficient database for Snapdeal that can organize and manage all essential business data in a secure and reliable manner. The database aims to minimize data redundancy, maintain data integrity, establish proper relationships between entities, and support fast data retrieval through SQL queries. This system will improve operational efficiency, simplify data management, and provide a strong foundation for the smooth functioning of the Snapdeal e-commerce platform.

Stakeholders

The stakeholders involved in the Snapdeal Database Project are:

- **Customers** – Purchase products, make payments, track orders, and provide ratings and reviews.
- **Sellers/Vendors** – List products, manage inventory, update prices, and process customer orders.
- **Administrators** – Manage users, products, categories, sellers, orders, and overall database operations.
- **Delivery Partners** – Handle shipment, delivery tracking, and order fulfillment.

- **Payment Gateway Providers** – Process online payments securely and maintain transaction records.
- **Customer Support Team** – Resolve customer complaints, manage returns, refunds, and order-related issues.
- **Database Administrators (DBAs)** – Design, maintain, secure, and optimize the database for efficient performance.
- **Business Management** – Monitor sales, customer activities, and business performance using reports and analytics for decision-making.

Business Requirements

- The system should maintain complete customer information, including registration and login details.
- The database should store and manage seller information and their product listings.
- The system should maintain a product catalog with categories, prices, stock availability, and product descriptions.
- Customers should be able to search, view, and purchase products easily.
- The database should record all orders, payments, shipping details, and delivery status.
- The system should support multiple payment methods, including online payments and Cash on Delivery (COD).
- The database should manage inventory by updating stock levels after every order.
- Customers should be able to submit product ratings and reviews.
- The system should support order cancellation, returns, and refund management.
- The database should generate reports on sales, customers, products, and seller performance to support business decision-making.
- The system should ensure secure storage of business and customer data while maintaining data accuracy and consistency.

Functional Requirements

- The system shall allow customers to register, log in, and manage their profiles.
- The system shall allow sellers to register, add, update, and remove product listings.
- The system shall store product details, including name, category, price, description, and stock quantity.

- The system shall enable customers to search, filter, and view product information.
- The system shall allow customers to add products to the shopping cart and place orders.
- The system shall record and manage order details, payment information, and delivery status.
- The system shall update inventory automatically after each successful order.
- The system shall support multiple payment methods, including online payments and Cash on Delivery (COD).
- The system shall allow customers to cancel orders, request returns, and track refunds.
- The system shall enable customers to submit product ratings and reviews.
- The system shall allow administrators to manage customers, sellers, products, categories, and orders.
- The system shall generate reports related to sales, inventory, customers, and seller performance.

Non-Functional Requirements

- The system should provide fast and efficient performance for data retrieval and transaction processing.
- The database should ensure high availability and reliability with minimal downtime.
- The system should maintain data security through user authentication and authorization.
- Customer and business data should be stored securely to protect privacy and prevent unauthorized access.
- The database should maintain data integrity by ensuring accurate and consistent information.
- The system should be scalable to support increasing numbers of customers, sellers, products, and orders.
- The database should support regular backup and recovery to prevent data loss.
- The system should be easy to maintain, update, and enhance as business requirements change.
- The application should provide a user-friendly interface for customers, sellers, and administrators.

- The system should support concurrent access by multiple users without affecting performance or data consistency.
- The database should be compatible with standard SQL-based database management systems and support future integration with other business applications.